

C: 399-A781MN477M1ELAS025TR-ND

T: IPD090N03LGATMA1TR-ND

D: RBR20NS30AFHTLTR-ND

L: 732-1171-2-ND

$$U_{in} = 48 \text{ V} \quad f = 1500 \text{ Hz}$$

$$U_{out} = 14 \text{ V}$$

$$i_{outmax} = 8 \text{ A}$$

$$\Delta i_L = 15\%$$

$$\Delta U_C = \Delta U_{out} = 200 \text{ mV}$$

PROJECT PART 1

$$\Delta i_L - \text{percentage of } i_{outmax} \Rightarrow \Delta i_L = \frac{15}{100} \cdot 8 = 1.20 \text{ A}$$

$$\mu = \frac{U_{out}}{U_{in}} = \frac{14}{48} = 0.29 \quad \Delta U_C = 0.200 \text{ V}$$

$$\Delta i_L = \left(\frac{U_{in} - U_{out}}{L} \right) \cdot \frac{\mu}{f} \Rightarrow L = \frac{(U_{in} - U_{out}) \cdot \mu}{\Delta i_L \cdot f}$$
$$= 5.505 \cdot 10^{-3} \text{ H} \approx 5.51 \text{ mH}$$

$$\Delta U_C = \frac{\Delta i_L}{8fC} \Rightarrow C = \frac{\Delta i_L}{\Delta U_C} \cdot \frac{1}{8f} = 5 \cdot 10^{-9} \text{ F} = 500 \text{ nF}$$

$$i_{savg} = (1 - \mu) \cdot i_{outmax} = (1 - 0.29) \cdot 8 \approx 5.64 \text{ A}$$

$$i_{Tmax} = i_{outmax} + \frac{\Delta i_L}{2} = 8 + \frac{1.20}{2} = 8.6 \text{ A}$$